Chapter 15

Repetitive Abnormalities

These are abnormal waveforms with relative uniform morphology that repeat either in rapid succession (rhythmic), after nearly regular intervals (periodic) or as recurring spike or sharp and slow waves (spike and wave). The pattern must continue for at least six cycles (six times) to qualify as a repetitive abnormality [1].

Therefore, repetitive abnormalities may be one of three types:

1. periodic;
2. rhythmic;
3. spike and wave.

Figure 15.1 shows sporadic and repetitive abnormalities.

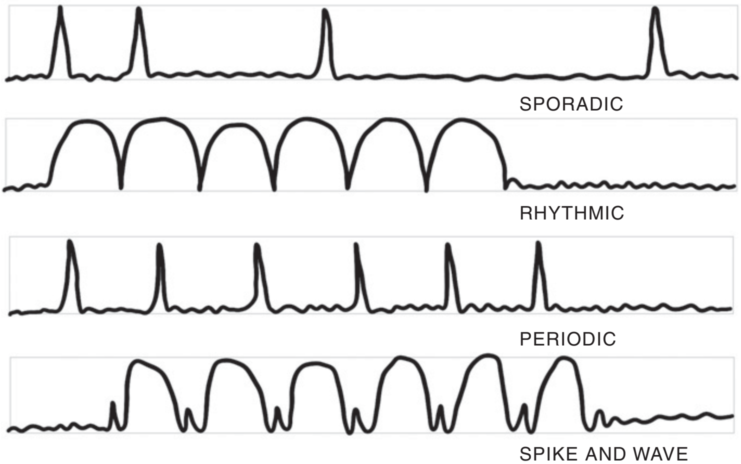

Figure 15.1 Sporadic and repetitive abnormalities.

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Repetitive abnormalities are common in critically ill patients and, like sporadic abnormalities, represent underlying cortical dysfunction. Periodic patterns are also markers of neuronal injury, independently predicting poor outcomes [1,2]. All repetitive abnormalities can be associated with epileptic seizures (epileptogenic); the strength of this association varies by abnormality. The association is maximum with lateralized periodic discharges (LPDs); it is also high with lateralized rhythmic delta activity (LRDA) and generalized periodic discharges (GPDs) but least with generalized rhythmic delta activity (GRDA). However, unlike sporadic abnormalities, repetitive abnormalities in specific situations may themselves represent the electrographic seizure activity, and these are called ictal patterns. This implies that repetitive abnormalities lie on an ictal–interictal continuum. An epileptogenic (interictal) pattern may evolve into an electrographic seizure (ictal pattern) and vice versa in the same patient during the same or subsequent recording. Therefore, determining the ictal potential of a repetitive pattern is crucial to its management [1,2,3]. Ictal patterns are described further in Chapter 16.

This chapter instructs the reader on how to correctly describe repetitive patterns based on standardized critical care EEG terminology proposed by the American Clinical Neurophysiology Society (ACNS) [1]. In the later part of this chapter, commonly encountered repetitive patterns are described along with potential ictal characteristics.

According to standardized terminology, the reader **should describe** each pattern as a combination of two main terms (main term 1 and main term 2). Subsequently, other descriptors, such as prevalence, duration, frequency, phases, sharpness, amplitude, and polarity, may be added to this combination. There is excellent interrater agreement on these main terms [1].

Main Term 1

Generalized (G). When a pattern equally involves both hemispheres, it is called a “generalized” pattern. It may show frontal, occipital, or central predominance based on its maximal amplitude in a referential montage.

Lateralized (L). When a pattern is prominent over one hemisphere, it is called a “lateralized” pattern. A lateralized pattern may have a focal or regional field, or it may be bilateral with a wide field but is clearly predominant over a single hemisphere.

When two distinct patterns of the same repetitive abnormality occur asynchronously, there are called independent patterns (I). The term *bilateral independent* (BI) is used when they occur bilaterally, and *unilateral independent* (UI) when they occur in the same hemisphere. The main term *multifocal*

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(Mf) is used to describe more than two distinct patterns of the same repetitive abnormality.

Main Term 2

Periodic Discharges (PD): These are repeating epileptic discharges with uniform morphologies that reappear at regular intervals. The reader should distinguish a discharge from a burst. Bursts have multiple phases (polyphasic) and longer durations (at least more than half a second).

Rhythmic Delta Activity (RDA): These are repeating monomorphic delta waves that recur uninterrupted, that is, without intervals between the consecutive waves.

Spike and Wave (SW): These are spike or sharp waves (or polyspikes) that are consistently followed by slow waves in a regular repeating run. Each spike or sharp wave is always followed by a slow wave such as a spike and wave, polyspikes and wave, or sharp and wave.

After you have closely inspected the repetitive pattern, choose an appropriate main term 1 and combine it with the appropriate main term 2 to describe the waveform. Example: A generalized periodic pattern would be called generalized (main term 1) periodic discharges (main term 2) or GPD for short.

If you think a pattern qualifies as both periodic and rhythmic, it should be coded as periodic with a rhythmic plus term (+R). A plus term renders the pattern ictal in appearance and is described in detail in Chapter 16.

All repetitive patterns including ictal patterns (seizures) may be induced by stimulation. The term *SI* is added before the combination of main terms to describe a stimulus induced pattern (e.g., SI-GPDs for stimulus-induced GPDs). The inducing stimulus should also be specified (example: suctioning). Stimulus-induced rhythmic, periodic, or ictal discharges (previously called SIRPIDs) are common in critically ill patients, but their true pathologic, prognostic, or therapeutic significance is not known [4]. Figure 15.2 shows a stimulus-induced GRDA (SI-GRDA).

Common Repetitive Abnormalities

Though many combinations of main terms 1 and 2 exist, the following abnormalities are commonly encountered in clinical practice:

Rhythmic Patterns

- generalized – rhythmic delta activity (GRDA)
- lateralized – rhythmic delta activity (LRDA)

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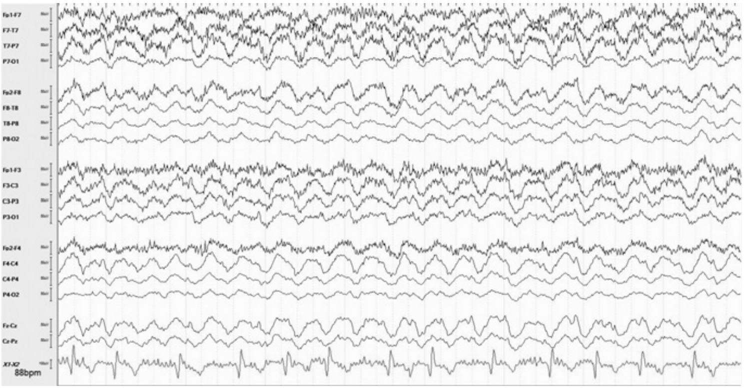

Figure 15.2 Eighty-three-year-old woman with dementia complicated by a urinary tract infection, stimulus-induced 2–3 Hz GRDA with tactile stimulation. There is prominent muscle artifact over the left hemispheric channels associated with arousal.

Periodic Patterns

- generalized – periodic discharges (GPD)
- generalized periodic discharges with triphasic morphology (triphasic waves)
- lateralized – periodic discharges (LPD)
- bilateral independent – periodic discharges (BIPD)

Spike and Wave Patterns

- generalized spike and wave (GSW)

Generalized Rhythmic Delta Activity (GRDA)

This is a diffuse, high-amplitude, monomorphic rhythmic delta activity that is commonly seen in encephalopathic patients. The pattern may be stimulus induced or associated with a state change (arousal) and is least associated with seizures. There are two common subtypes:

Frontally Predominant GRDA (Common in Adults): This was previously called frontal intermittent rhythmic delta activity (FIRDA). This pattern is nonspecific for etiology and is least associated with epileptic seizures. It is most commonly associated with toxic or metabolic encephalopathies, although it can also be seen with structural brain lesions or raised intracranial pressure [5]. Figure 15.3 shows frontally predominant GRDA.

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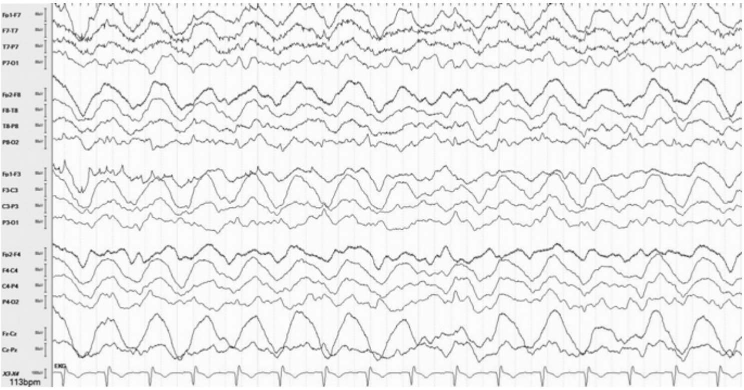

Figure 15.3 Sixty-five-year-old man with sepsis, frontally predominant 1.5–2 Hz GRDA.

Occipitally Predominant GRDA (Common in Children): This was previously called occipital intermittent rhythmic delta activity (OIRDA) and is a nonspecific pattern, but rarely it is associated generalized epilepsies (epileptogenic), particularly childhood absence epilepsy (CAE). It is more commonly associated with a toxic metabolic encephalopathy, structural brain lesions, or raised intracranial pressure [6]. Figure 23.3 (Chapter 23) shows occipital predominant GRDA (or OIRDA) in a patient with childhood absence epilepsy (CAE).

Lateralized Rhythmic Delta Activity (LRDA)

This is a focal monomorphic, often sharply contoured, rhythmic delta slowing that occurs in short bursts of a few seconds but may be longer in encephalopathic patients. The pattern is most commonly predominant over the temporal region but is not restricted to it. LRDA is independently associated with epileptic seizures (epileptogenic). Focal abnormalities such as LPDs often coexist. Typically, an underlying acute or remote structural lesion is present [7]. Temporal intermittent rhythmic delta activity (TIRDA) is a subtype of LRDA that is seen in patients with mesial temporal lobe epilepsy, typically seen in the ambulatory setting [8]. Figure 15.4 shows left frontal–predominant LRDA, whereas Figure 20.2 (Chapter 20) shows left temporal intermittent rhythmic delta activity (TIRDA).

Generalized Periodic Discharges (GPD)

These are diffuse, usually symmetric, periodic discharges. They may show frontal or occipital predominance. They were previously called generalized

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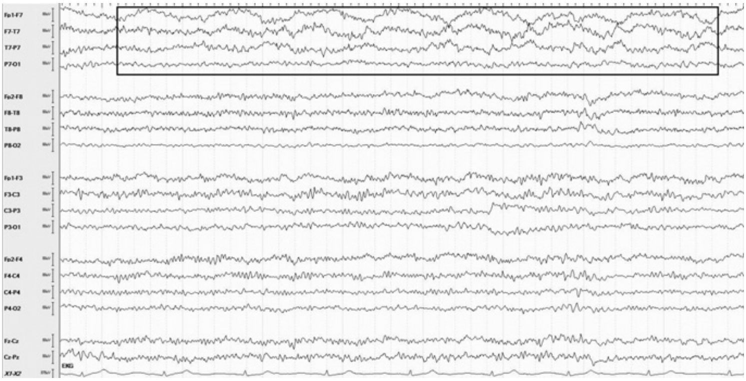

Figure 15.4 Sixty-two-year-old woman with a first-time seizure, 1 Hz left frontal–predominant LRDA.

periodic epileptiform discharges (GPEDs), but the word *epileptiform* has since been removed as they may not always be associated with seizures (e.g., nonictal triphasic waves) and their morphology isn't always epileptiform. GPDs may be stimulus induced (SI-GPDs) or associated with a change of state (e.g., arousals). In specific situations, GPDs may represent the electrographic manifestation of electrographic seizures and therefore constitute an ictal pattern.

GPDs are commonly seen in the following conditions:

1. toxic or metabolic encephalopathies, including sepsis (esp. triphasic waves);
2. extensive structural cerebral damage such as hypoxic ischemic encephalopathy, diffuse intracranial hemorrhage, or neurodegeneration;
3. epileptic encephalopathies;
4. medications such as cefepime, Lithium, Ifosfamide, Baclofen, or Metrizamide;
5. infections such as Creutzfeldt–Jakob disease (CJD) or subacute sclerosing panencephalitis (SSPE).

GPDs are traditionally thought to imply poor prognosis, however, recent studies have not found an independent association with increased mortality. GPDs have been associated with nonconvulsive status epilepticus (NCSE), which is itself associated with increased mortality. The underlying etiology and severity of the primary disease process are most often the key determinants of prognosis in these patients [9,10]. Figure 15.5 shows occipital-predominant GPDs.

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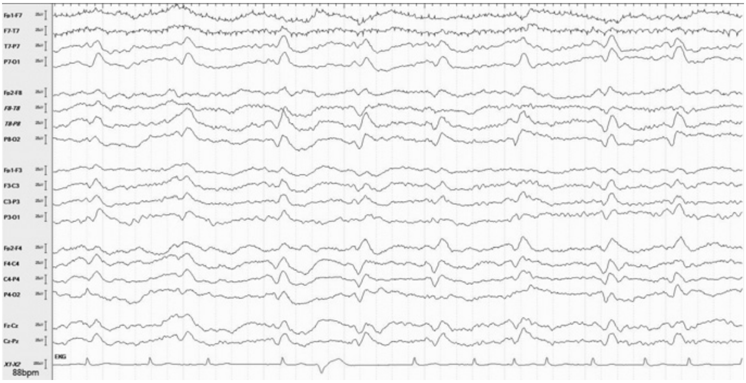

Figure 15.5 Seventy-five-year-old man with Creutzfeldt–Jakob disease, 1 Hz occipital-predominant GPDs.

Generalized Periodic Discharges with Triphasic Morphology (Triphasic Waves)

This is a common subtype of GPDs that is encountered in patients with toxic or metabolic encephalopathies (such as hepatic or uremic encephalopathies) but may also be seen with multiple different etiologies including hypoxic and epileptic encephalopathies. Less commonly, triphasic waves may be the electrographic manifestation of NCSE. These ictal triphasic waves can be distinguished from nonictal triphasic waves based on electrographic features and a response to trials of benzodiazepines or nonsedating antiepileptic medications such as levetiracetam [11].

Triphasic waves (TW) are identified on EEG using two key features:

1. Triphasic waves often have two baseline crossings or three phases. “Phases” are determined by the number of baseline crossings seen in a typical discharge in longitudinal bipolar montage. Look at the channel in which the discharge is best appreciated. A triphasic wave will typically cross the baseline twice, each of the three phases is longer than the previous and may be negative (up)-positive(down)-negative(up). The positive phase (down/phase II) is typically of highest amplitude. Sometimes, the initial negative phase is not clearly seen and only a prominent positive (down) and negative (up) phase are seen, these bi-phasic waves (phases II–III only) may also be categorized as triphasic waves [1].

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Figure 15.6 Seventy-seven-year-old man with sepsis on cefepime, 1–2 Hz triphasic waves with prominent AP phase lag.

2. Triphasic waves often have an anterior–posterior or posterior-anterior lag. This applies if a consistent measurable delay of more than 100 ms exists from the anterior most channel to the posterior most channel in which the wave is seen or vice versa (posterior-anterior lag). You should use both a longitudinal and a referential montage (ipsilateral ear reference) to evaluate lag [1]. Figure 15.6 shows generalized periodic discharges with triphasic morphology (triphasic waves).

After you have identified triphasic waves, the next step is to estimate if they are potentially ictal. This can be done based on electrographic features and a response to anti-epileptic medication trial.

Electrographic Features

Electrographic features suggestive of potentially nonictal triphasic include the following:

- presence of a phase lag;
- dominant phase II (down);
- longer duration of phase I (not a sharp/spike);
- slower frequencies;
- stimulus induction or with arousal from drowsiness/sleep (state dependency).

Features suggestive of potentially ictal triphasic include the following:

- overriding spikes or sharp waves;
- faster frequencies (>2.5 Hz);
- no change in the pattern with state change or stimulation.

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However, it must be noted that no specific electrographic features have been identified that either alone or in combination can reliably distinguish triphasic waves that respond to treatment (presumably ictal) vs. those that do not (presumably nonictal). Therefore, experts have proposed a trial of anticonvulsant medications in cases of incompletely explained encephalopathies with triphasic waves. Based on a single retrospective study, almost one-third of all "metabolic" triphasic waves may show a definitive response to medication [11,12].

Anti-epileptic Medications Trial

Ictal triphasic waves are thought to respond to anti-epileptic medications (responders). Trials of low-dose benzodiazepines or a nonbenzodiazepine anti-epileptics (e.g., levetiracetam) may be used to evaluate for a response. Typically, the patient is monitored at least for 24–48 hours with continuous electroencephalography during the trial.

Low-Dose Benzodiazepine Trial

One method is to administer small doses of lorazepam (up to 4 mg in divided doses of 1 mg each) with repeated clinical and EEG assessment before the first dose and after each subsequent dose. A successful trial is defined as the resolution of the triphasic waves AND either clear clinical improvement or appearance of previously absent normal EEG patterns (e.g., posterior-dominant rhythm). Mere resolution of the EEG without clinical improvement is considered an equivocal response. Most responses occur immediately (within 2 hours of initiating lorazepam), but late responders have also been described. Further doses of lorazepam should not be administered in the setting of hypotension or respiratory depression [11,12].

Levetiracetam Trial

Recent studies have shown a greater response rate of triphasic waves to non benzodiazepines anti-epileptics (most commonly levetiracetam), compared to benzodiazepines, suggesting this to be a useful strategy even in those with an inconclusive response to benzodiazepines. Late responses were also common (greater than 2 hours after initiating the trial). A loading dose of levetiracetam (25 mg/kg as a single intravenous dose) followed by maintenance dosing of 1.5 gram twice daily for 48 hours is typically used in our practice. Success of the trial is defined in the same way as for lorazepam that is, resolution of the triphasic waves AND either definite clinical improvement or emergence of previously absent normal patterns such as a posterior dominant rhythm. Other less sedating antiepileptics may also be used in place of levetiracetam [11,12,13]. Figure 15.7 shows apparent ictal triphasic waves.

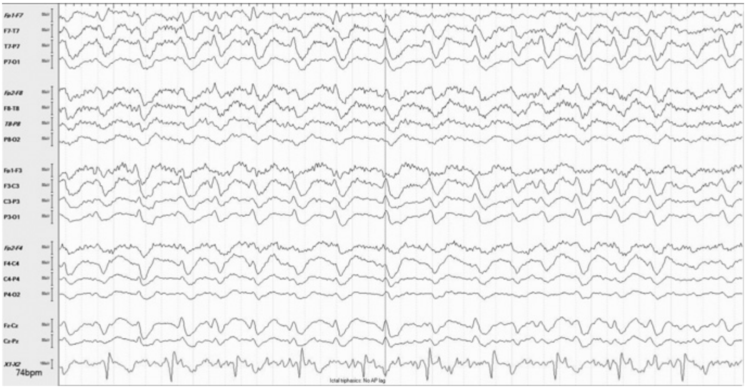

Figure 15.7 Same patient as in Figure 15.3, 2–2.5 Hz triphasic waves without AP phase lag.

Lateralized Periodic Discharges (LPD)

These are focal periodic patterns, lateralized to a single hemisphere (sometimes with wide fields). Previously, they were called periodic lateralized epileptiform discharges (PLEDs). They commonly lie close to the ictal end of the ictal–interictal continuum and in specific cases represent an ictal pattern. LPDs represent an underlying acute destructive cortical process that may not always be evident on neuroimaging. They appear within the first few days of injury and usually resolve in a few weeks. Chronic LPDs suggest the development of focal epilepsy. All forms of structural damage may cause LPDs; strokes, viral encephalitis, and hypoxic injury being the most common etiologies. LPDs are strongly associated with epileptic seizures (as high as 60% per some studies) [14,15]. Focal motor seizures being the commonest seizure type [14]. Seizures may appear late and they may not be convulsive. Therefore, continuous EEG monitoring for 24–72 hours should be considered in those with LPDs [16]. LPDs are also an independent risk factor for poor outcome in those with SAH and ICH [17,18]. Figure 15.8 shows right hemispheric LPDs.

After you have identified LPDs, the next step is to determine if they are ictal. Ictal LPDs can be distinguished from interictal LPDs based on their clinical correlates and electrographic features [19]:

Common Clinical Correlates of Ictal LPDs

1. *Focal clonic seizures (or epilepsia partialis continua):* Focal seizures or status epilepticus may be seen contralateral to the LPDs. Often, the

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Figure 15.8 Seventy-two-year-old lady with HSV encephalitis, right hemispheric LPDs.

individual discharges are time locked to the EMG recording or myogenic artifact.

2. Subtle bedside clinical signs such as eye deviation, aphasia, hemiparesis, hemianopia, sensory changes, or confusion may be associated with LPDs located over eloquent areas such as fronto-polar, temporo-parietal and occipital regions. In these cases, a resolution of the LPDs in addition to clinical improvement (focal symptoms or encephalopathy) in response to a low dose of lorazepam or loading dose of antiepileptic medication may suggest ictal LPDs. Further escalation of antiepileptic therapy may be necessary.
3. Some experts have suggested that evidence of regional hypermetabolism such as an increased signal on PET or SPECT imaging with LPDs supports an ictal pattern and warrants aggressive treatment [20].

The electrographic features of ictal LPDs are described later in Chapter 16.

Bilateral Independent Periodic Discharges (BIPD)

These are bilateral, independent, often asynchronous periodic discharges. They are less common compared to LPDs but with similar etiology. They are also thought to be associated with an increased seizure risk and are often associated with severe encephalopathy [21]. Figure 15.9 shows BIPDs that evolve into a seizure.

Generalized Spike and Wave (GSW)

Repetitive generalized spike and wave (GSW) discharges are epileptiform patterns, and those occurring in runs of greater than 2.5 Hz are potentially

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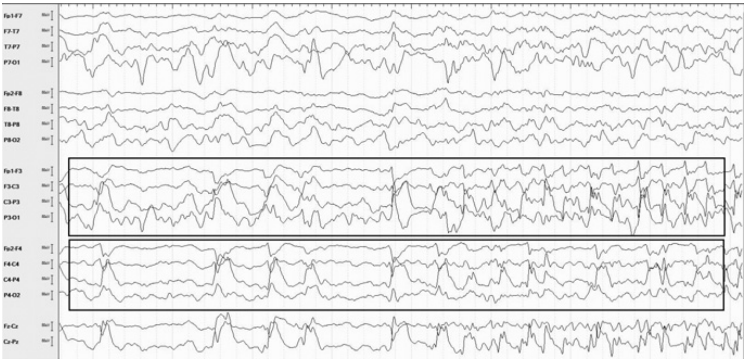

Figure 15.9 Sixty-year-old man with hepatic coma, BIPDs (central predominant) that evolve into an electrographic seizure over the left paracentral region.

ictal. Complexes may be composed of polyspikes/sharps and waves [1]. These patterns are further described in Chapter 16.

Chapter Summary

1. Repetitive waveforms may repeat in quick succession (rhythmic), after nearly regular intervals (periodic), or as spike and wave (SW).
2. Periodic discharges are considered markers for neuronal injury.
3. Repetitive abnormalities are epileptogenic, but the strength of their association with seizures is variable; in specific situations they may also represent ongoing electrographic seizures (ictal patterns).
4. A repetitive pattern is described as a combination of a main term 1 with a main term 2 based on the American Clinical Neurophysiology Society (ACNS) standardized critical care EEG terminology.
5. Main term 1 could be either generalized (G), lateralized (L), bilateral or unilaterally independent (BI or UI), or multifocal (Mf).
6. Main term 2 could be either periodic discharges (PD), rhythmic delta activity (RDA), or spike and wave (SW).
7. Common rhythmic abnormalities include GRDA and LRDA. Common periodic abnormalities include GPDs, LPDs, and BIPDs.
8. The term *SI* is added to the combination of main terms to denote a stimulus-induced pattern (e.g., SI-GPDs for stimulus-induced GPDs).
9. Frontal-predominant GRDA are most commonly seen in adults with toxic or metabolic encephalopathies, including sepsis. They have the least association with seizures.

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10. Occipital-predominant GRDA are common in children and may rarely be associated with a generalized epilepsy syndrome (e.g., childhood absence epilepsy).
11. LRDA is an epileptogenic pattern seen in critically ill patients with structural lesions; TIRDA is a subtype of LRDA associated with mesial temporal lobe epilepsy.
12. GPDs are a common pattern in critically ill patients associated with nonconvulsive seizures. Triphasic waves are a subtype of GPDs with triphasic morphology and usually an anterior–posterior or less commonly a posterior–anterior lag.
13. Triphasic waves may be potentially ictal or nonictal. These can be distinguished based on their electrographic features and response to a trial of antiepileptic medications.
14. LPDs are highly epileptogenic patterns and represent the electrographic markers of an acute destructive process of the underlying cortex.
15. Ictal LPDs can be identified based on their clinical accompaniments, regional increase in cerebral metabolism, and electrographic features.
16. BIPDs are bilateral PDs and are usually asymmetric. They are less common compared to LPDs and are usually seen in severe encephalopathy.
17. GSW discharges of more than 2.5 Hz are potentially ictal patterns.
18. Determining the ictal potential of a repetitive pattern is the key to its appropriate management.

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